ISO Standard-Based Energy Audit and 10-Year Energy Demand Forecasting for Barangay Lower Bicutan, Taguig City, Philippines

C-6, Taguig, 1632 Metro Manila, Philippines

In Partial Fulfillment of the Requirements for the Course
Elective 1: Renewable Energies

by

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Submitted to
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TABLE OF CONTENTS

Title Page i
Table of Contents ii

A. BACKGROUND 1
a. General Information of Audited of Audited Organization 1
b. Occupancy and Use Schedules 1
c. Statement of Confidentiality (Limitations) 2
B. ENERGY PROFILE OF BARANGAY 2
a. Energy Sources 2
b. Energy Bill Analysis 2
C. ENERGY DEMAND ANALYSIS 4
a. Forecast Methodology 4
b. Forecast Results 5
D. ENERGY AUDIT 7
a. Site Inspection Findings 7
b. Energy Performance Analysis 18
c. Evaluation of Improvement Opportunities 18
d. Recommendations for Improvement of Energy Performance 21
e. Cost-Savings Analysis 24
E CONCLUSIONS 26
a. Summary of Findings and Recommendations 26
b. Next Steps 27
F REFERENCES 27
G APPENDICES 28
a. Code 28
b. Documentation 31
c. Monthly Bill 35

A. BACKGROUND

a) General Information of Audited of Audited Organization



Figure 1: *Lower Bicutan Barangay Hall Building.*

Taguig is a landlocked highly urbanized city in the National Capital Region. The city is divided into 28 barangays, one of which is Barangay Lower Bicutan, situated in District 2 [1].

The Barangay building is a three-story structure painted beige with maroon on the roofing. It shares its location with a Sangguniang Kabataan (SK) Building, though the two functions are distinct and separate. For the energy audit, only the barangay building was assessed, not the SK Building. Due to security reasons, a floor plan was not available, but upon site inspection, the building features thirteen (13) rooms, a gym, a multi-purpose hall and a therapy room.

b) Occupancy and Use Schedules

The barangay hall is open from 8 AM to 5 PM on weekdays, staffed by barangay staff members and officers. Additionally, the security room are operational 24 hours a day, seven days a week.

c) Statement of Confidentiality / Limitations

This energy audit was composed and examined by the specified auditors as a component of the fourth-year curriculum requirement for the Elective: Renewable Energies. The project is titled " ISO Standard-Based Energy Audit and 10-Year Energy Demand Forecasting for Barangay Lower Bicutan, Taguig City, Philippines." In line with a verbal agreement with barangay officials, all information provided by them, along with data gathered by the auditors, will be exclusively used to meet the requirements of this academic project. We are committed to not disclosing any information that is not essential for completing the project. Following the analysis, the team will present their findings and propose energy efficiency recommendations to the barangay officials.

Fortunately, the auditors accessed all the rooms including the security room full of CCTV cameras and the barangay chairman’s office, taking pictures, recording videos, interviews and taking notes throughout their inspection.

B. ENERGY PROFILE OF BARANGAY

a) Energy Sources

The Manila Electric Company (MERALCO) is the largest electric distribution utility in the Philippines' private sector, serving 39 cities and 72 municipalities. Although its franchise area spans just over 9,685 km², which is only 3% of the Philippines' total land area, it delivers 55% of the nation's electricity output [2]. Among the areas it serves is Taguig City, where Barangay Lower Bicutan is located. The barangay building is categorized under Meralco's General Service B (GS-B) business customer type, which is applicable to non-residential customers with a contracted capacity between 5 kW and 39 kW [3].

b) Electric Bill Analysis

i. Monthly Electricity Consumption (kWh)

Table 1 presents key details, including the year, billing date, monthly electricity rate in pesos, and the corresponding actual consumption based on the barangay's electric bills. Figure 2 complements this by visualizing the data in graphical form, illustrating trends and variations in electricity usage. The data was sourced directly from photocopied electric bills provided by the barangay, covering the period from February 2022 up to November 2024

Breakdown of Energy Consumption of Barangay Lower Bicutan			
Year	**Billing Date**	**Monthly Rate (Peso)**	**Actual Consumption (kWh)**
2022	20-02-2022	9.79	4904
2022	21-03-2022	9.78	4820
2022	21-04-2022	9.93	6439
2022	21-05-2022	10.25	5537
2022	19-06-2022	10.32	6380
2022	20-07-2022	10.08	5848
2022	19-08-2022	9.99	5628
2022	19-09-2022	10.55	5176
2022	20-10-2022	10.19	5744
2022	19-11-2022	10.15	6102
2022	19-12-2022	10.51	5625
2023	20-01-2023	11.29	5080
2023	20-02-2023	11.01	5379
2023	22-03-2023	11.62	4985
2023	21-04-2023	11.21	5974
2023	20-05-2023	11.27	6577
2023	19-06-2023	11.46	5635
2023	20-07-2023	10.77	5805
2023	20-08-2023	10.89	4813
2023	20-09-2023	11.51	4517
2023	20-10-2023	11.88	4529
2023	20-11-2023	12.14	4616
2023	20-12-2023	11.54	4129
2024	21-01-2024	12.05	3540
2024	19-02-2024	12.15	4354

2024	20-03-2024	12.38	4439
2024	29-04-2024	11.54	3993
2024	24-05-2024	11.00	5680
2024	27-06-2024	9.07	5784
2024	25-07-2024	11.51	4886
2024	21-08-2024	11.04	6168
2024	21-09-2024	11.63	5346
2024	23-10-2024	11.58	4859
2024	22-11-2024	11.85	4875

Table 1: *Monthly Energy Consumption of Lower Bicutan Barangay from February 2022-November 2024*

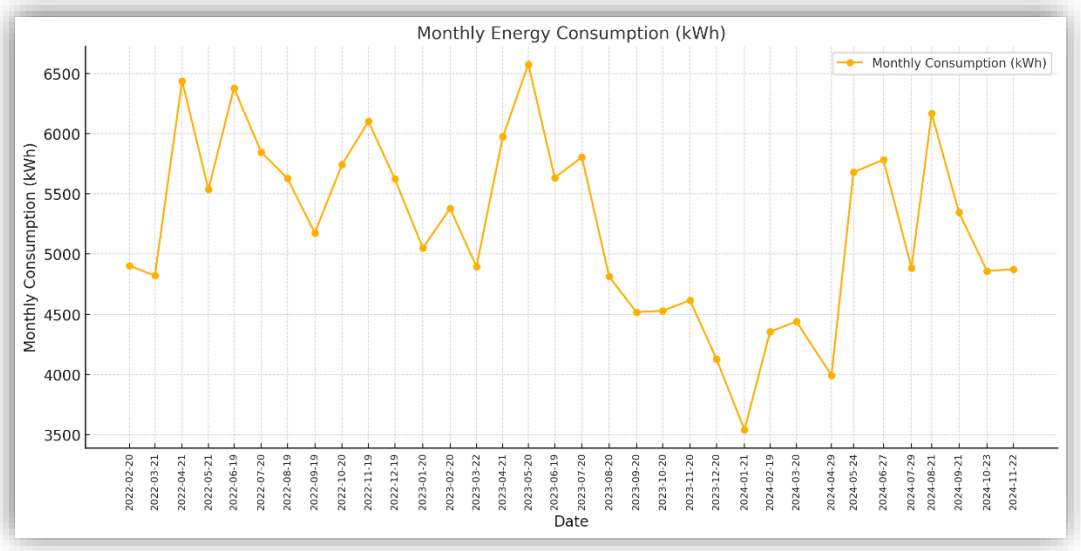


Figure 1: *Graph of Monthly Electricity Consumption*

C. ENERGY DEMAND ANALYSIS

a. Forecast Methodology

Forecasting is the process of predicting future trends or events based on historical data, enabling organizations to make informed, proactive decisions. In energy audits, accurate forecasts assist in estimating future energy demands, identifying inefficiencies, and planning sustainable resource management strategies. By

leveraging forecasting, organizations can optimize resource allocation, reduce costs, and enhance sustainability efforts [4].

Prophet, developed by Facebook, is a widely used forecasting model known for its ability to handle time-series data with complex seasonal patterns, trends, and outliers. It employs an additive model framework.

$$y(t) = g(t) + s(t) + h(t) + \in_t$$

where $y(t)$ is the forecasted value at time *t*. The components of this model include $g(t)$ which represents the trend capturing long-term growth or decline. $s(t)$ which accounts for periodic variations such as seasonal cycles. $h(t)$ which incorporates holiday effects or specific events influencing the data. Lastly, $\in_t$ the error term accounting for unexplained variations.

This model suitable for energy audits where consumption patterns often exhibit regular seasonal and daily variations influenced by factors like weather and operations [5]. Prophet's flexibility allows auditors to model long-term trends effectively, accommodating periodic fluctuations and unexpected changes, which is essential for generating accurate 10-year energy forecasts.

In this energy audit, the auditor used Jupyter Notebook due to its interactive and flexible environment that combines code, outputs, and documentation in a single platform. Jupyter's support for Python, the programming language used to implement Prophet, makes it a suitable choice for executing, visualizing, and iterating forecasting models. Additionally, it fosters reproducibility and collaboration, ensuring that the results are well-documented and easily shared among stakeholders [6].

b. Forecast Results

This section outlines the anticipated energy demand for the barangay over the upcoming decade. Figure 3 showcases the 10-year forecast of energy consumption, emphasizing both the underlying trends and seasonal fluctuations. Table 2 presents a comprehensive breakdown, listing each year alongside its respective peak energy demand (kWh) from 2025 to 2035.

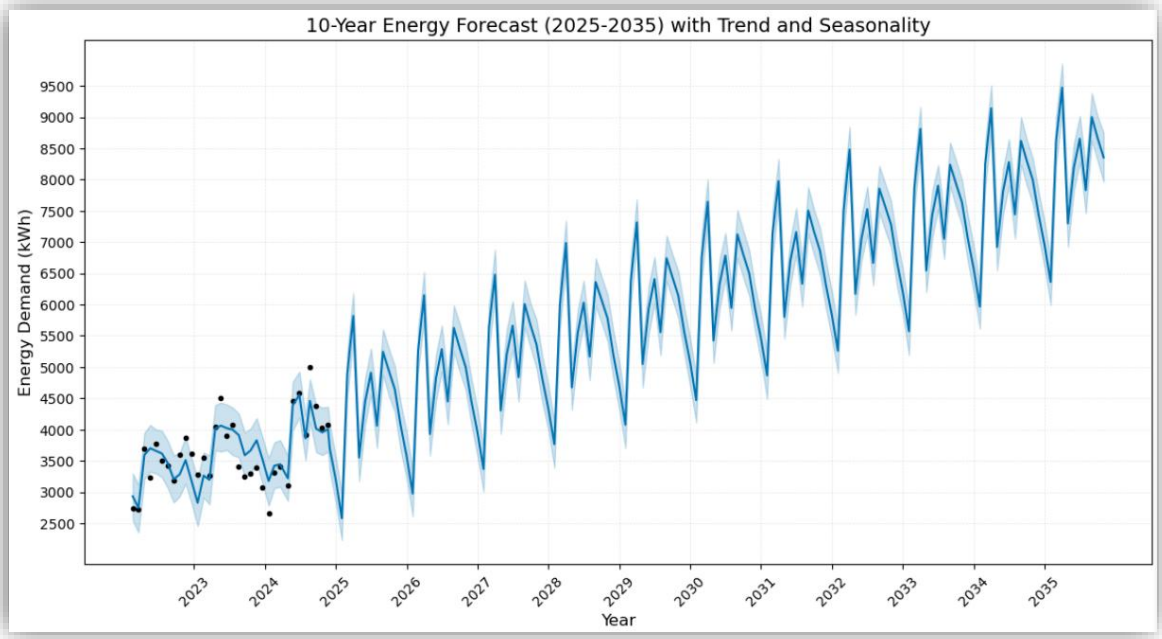


***Figure 3**: "10-Year Peak Energy Demand Forecast (2025–2035)"*

YEAR	Peak Energy Demand (kWh)
2025	5818.901
2026	6148.894
2027	6478.007
2028	6984.674
2029	7315.610
2030	7645.603
2031	7974.716
2032	8481.383
2033	8812.319
2034	9142.312
2035	9471.425

***Table 3**: "Forecasted Peak Energy Demand by Year (2025–2035)"*

The energy demand forecast shows a rapid upward trajectory, starting at 5,531.946 kWh in 2025 and reaching 9,641.515 kWh by 2035. This accelerated growth of

nearly 4,110 kWh over 10 years reflects an average annual increase of approximately 7.4%, indicating a strong correlation between economic expansion and energy needs. Seasonal variations within each year remain consistent, but the overall trend demonstrates that higher demand levels are becoming the new baseline. By 2035, the forecast nears the critical threshold of 10,000 kWh, emphasizing the urgency of addressing capacity constraints in the energy system.

D. ENERGY AUDIT

a) Site Inspection Findings

i. Lighting Systems

Breakdown of Energy Consumption by Usage and Source										
Fixture/Model	Quantity	Watt Rating (W)	Est. Hours Used per Day	Hours	Days	Location	Number of Working	Number of Not Working	Monthly Consumption (kWh)	Surveyed By:
Lighting System										
2x28-2x36 watt Flourescent Grill Surface Light	4	28	8 hours (5 days per week)	8	22	Clearance Section Room	3	1	14.784	Dran
2x28-2x36 watt Flourescent Grill Surface Light	2	28	8 hours (5 days per week)	8	22	1st Floor Hallway	2	0	9.856	Dran
2x28-2x36 watt Flourescent Grill Surface Light	3	28	8 hours (5 days per week)	8	22	Information Desk	3	0	14.784	Dran

2x28-2x36 watt Flourescent Grill Surface Light	2	28	10 hours (7 days)	10	30	Emergency Medical Service Room	2	0	16.8	Dran
LED light bulb	1	8	10 hours (7 days)	10	30	Emergency Medical Service Alert Room	1	0	2.4	Dran
LED light bulb	4	8	10 hours (7 days)	10	30	Barangay Security Force Room	4	0	9.6	Dran
Flourescent Light Tube	1	28	10 hours (7 days)	10	30	Office of the Investigation	1	0	8.4	Dran
2x28-2x36 watt Flourescent Grill Surface Light	7	28	10 hours (7 days)	10	30	Office of the Investigation	7	0	58.8	Dran
2x28-2x36 watt Flourescent Grill Surface Light	6	28	8 hours (5 days per week)	8	22	Annex Building	2	4 are only optional to use	9.856	Dran
LED light bulb	2	8	2 hours (5 days per week)	2	22	Annex Building (Comfort Room)	2	0	0.704	Dran
LED light bulb	3	8	8 hours (5 days per week)	8	22	Office of the Secretary	3	0	4.224	Dran

2x28-2x36 watt Flourescent Grill Surface Light	2	28	8 hours (5 days per week)	8	22	Barangay Captain's Room	2	0	9.856	Dran
LED light bulb	6	8	2 hours (5 days per week)	2	22	Barangay Captain's Room	6	0	2.112	Dran
2x28-2x36 watt Flourescent Grill Surface Light	2	28	8 hours (5 days per week)	8	22	2nd Floor Hallway	2	0	9.856	Dran
LED light bulb	1	8	8 hours (7 days per week)	8	30	CCTV Control Room	1	0	1.92	Dran
2x28-2x36 watt Flourescent Grill Surface Light	4	28	8 hours (5 days per week)	8	22	Office of Barangay Justice System Room	4	0	19.712	Dran
2x28-2x36 watt Flourescent Grill Surface Light	2	28	8 hours (5 days per week)	8	22	Office of Land Dispute Room	2	0	9.856	Dran
2x28-2x36 watt Flourescent Grill Surface Light	4	28	8 hours (5 days per week)	8	22	Session Hall	4	0	19.712	Dran

2x28-2x36 watt Flourescent Grill Surface Light	15	28	Dependent on events schedule			3rd Floor	15	0	21	Dran
LED light bulb	1	8	8 hours (5 days per week)	8	22	Treasury Department	1	0	1.408	Dran
2x28-2x36 watt Flourescent Grill Surface Light	4	28	8 hours (5 days per week)	8	22	Treasury Department	4	0	19.712	Dran

ii. Cooling Equipments

Breakdown of Energy Consumption by Usage and Source										
Fixture/Model	Quantity	Watt Rating (W)	Est. Hours Used per Day	Hours	Days	Location	Number of Working	Number of Not Working	Monthly Consumption (kWh)	Surveyed By:
HVAC system										
Fujidenzo Floor Standing Aircon	1	1445	8 hours (5 days per week)	8	22	Clearance Section Room	1	0	254.32	Dran
Samsung Aircon	1	523.2	8 hours (7 days per week)	8	30	Emergency Medical Service Room	1	0	125.568	Dran

Fujidenzo Floor Standing Aircon	1	1445	8 hours (7 days per week)	8	30	Office of the Investigation	1	0	346.8	Dran
Koppel Floor Standing Aircon	2	850	8 hours (5 days per week)	8	22	Annex Building	1	1 not in use	149.6	Dran
Split Type Aircon	1	320	8 hours (7 days per week)	8	30	Barangay Captain's Room	1	0	76.8	Dran
Split Type Aircon	2	320	24 hours (7 days per week)	24	30	CCTV Control Room	2	0	460.8	Dran
Split Type Aircon	1	320	8 hours (5 days per week)	8	22	Office of Barangay Justice System Room	1	0	56.32	Dran
Split Type Aircon	1	320	8 hours (5 days per week)	8	22	Office of Land Dispute Room	1	0	56.32	Dran
Koppel Floor Standing Aircon	1	850	8 hours (5 days per week)	8	22	Session Hall	1	0	149.6	Dran
Carrier Aircon	1	437.5	8 hours (5 days per week)	8	22	Treasury Department	1	0	77	Dran

Hanabishi Industrial Wall Electric Fan	1	180	Half an hour (2 days per week)	0.5	10	Clearance Section Room	1	0	0.9	Dran
Hanabishi Industrial Orbit Fan	1	75	Half an hour (2 days per week)	0.5	10	Clearance Section Room	1	0	0.375	Dran
Dowell Stand Fan	1	60	Half an hour (2 days per week)	0.5	10	Clearance Section Room	1	0	0.3	Dran
3D Mini Fan	1	50	Half an hour (2 days per week)	0.5	10	Clearance Section Room	1	0	0.25	Dran
Tornado BLASTER 06B	1	350	8 hours (5 days per week)	8	22	Information Desk	1	0	61.6	Dran
Stand Fan	1	60	Half an hour (3 days per week)	0.5	14	Emergency Medical Service Room	1	0	0.42	Dran
Industrial Orbit Fan	3	75	Half an hour (3 days per week)	0.5	14	Office of the Investigation	3	0	1.575	Dran
Industrial Orbit Fan	3	75	1 hour (4 days per week)	1	18	Annex Building	3	0	4.05	Dran

Tornado BLASTER 06B	1	150	8 hours (working days)	8	22	Annex Building	0	1 not in use	0	Dran
Stand Fan	1	60	Half an hour (5 days per week)	0.5	22	Office of Barangay Justice System Room	1	0	0.66	Dran
Industrial Orbit Fan	2	75	Half an hour (5 days per week)	0.5	22	Session Hall	2	0	1.65	Dran

iii. Other Systems/Auxilliary Appliances

Breakdown of Energy Consumption by Usage and Source										
Fixture/Model	Quantity	Watt Rating (W)	Est. Hours Used per Day	Hours	Days	Location	Number of Working	Number of Not Working	Monthly Consumption (kWh)	Surveyed By:
Auxiliary Systems										
Devant 28DL420 28-inch LED TV	1	55	1 hour (2 days per week)	1	10	Clearance Section Room	1	0	0.55	Dran
Kyocera Printer	1	76	An hour (1 day per week)	1	5	Clearance Section Room	1	0	0.38	Dran
Personal Computer	3	115	8 hours (5 days per week)	8	22	Clearance Section Room	3	0	60.72	Dran

CCTV Camera	2	4	8 hours (5 days per week)	8	22	Clearance Section Room	2	0	1.408	Dran
Water Dispenser	1	250	8 hours (5 days per week)	8	22	Clearance Section Room	1	0	44	Dran
Speaker	1	20	2 hours (2 days per week)	2	10	Clearance Section Room	1	0	0.4	Dran
Coffee Vending Machine SC 8703	1	150	8 hours (3 days per week)	8	14	Information Desk	1	0	16.8	Dran
Rice Cooker	1	200	Half an hours (3 days)	0.5	14	Emergency Medical Service Room	1	0	1.4	Dran
Water Dispenser	1	250	8 hours (5 days per week)	8	22	Office of the Investigation	1	0	44	Dran
TCL TV	1	30	1 hour (1 day per week)	1	5	Annex Building	1	0	0.15	Dran
Personal Computer	3	115	8 hours (5 days per week)	8	22	Annex Building	3	0	60.72	Dran

Kyocera Printer	1	76	An hour (1 day per week)	1	5	Office of the Secretary	1	0	0.38	Dran
Epson Printer	2	10	2 hours (3 days per week)	2	14	Office of the Secretary	2	0	0.56	Dran
Inverter type Refrigerator	1	32	24 hours (5 days per week)	24	22	Office of the Secretary	1	0	16.896	Dran
Personal Computer	4	115	8 hours (5 days per week)	8	22	Office of the Secretary	4	0	80.96	Dran
TCL TV	1	30	2 hours (2 days per week)	2	10	Office of the Secretary	1	0	0.6	Dran
Internet Router	3	20	8 hours (5 days per week)	8	22	Office of the Secretary	3	0	10.56	Dran
CCTV Monitor	1	30	24 hours (5 days per week)	24	22	Barangay Captain's Room	1	0	15.84	Dran
CCTV Camera	300	4	24 hours (7 days per week)	24	30	Spread around Barangay Lower Bicutan	300	0	864	Dran

CCTV Monitor	13	30	24 hours (7 days per week)	24	30	CCTV Control Room	13	0	280.8	Dran
KEVLER PA-500 PA Power Amplifier	1	450	24 hours (7 days per week)	24	30	CCTV Control Room	1	0	324	Dran
Kevler BGM-500UB Power Amplifier	2	450	24 hours (7 days per week)	24	30	CCTV Control Room	2	0	648	Dran
Kevler BGM-500 Power Amplifier	1	450	24 hours (7 days per week)	24	30	CCTV Control Room	1	0	324	Dran
Personal Computer	2	115	8 hours (5 days per week)	8	22	Office of Barangay Justice System Room	2	0	40.48	Dran
Epson Printer	1	10	2 hours (3 days per week)	2	14	Office of Barangay Justice System Room	1	0	0.28	Dran
CCTV Camera	1	4	24 hours (5 days per week)	24	22	Office of Barangay Justice System Room	1	0	2.112	Dran
Epson Printer	1	10	2 hours (3 days per week)	2	14	Office of Land Dispute Room	1	0	0.28	Dran
Personal Computer	1	115	8 hours (5 days per week)	8	22	Office of Land Dispute Room	1	0	20.24	Dran

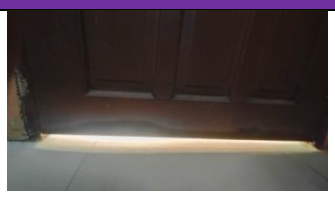
CCTV Camera	2	4	24 hours (5 days per week)	24	22	Session Hall	2	0	4.224	Dran
TCL TV	1	30	3 hours (5 days per week)	3	22	Session Hall	1	0	1.98	Dran
Ceragem	1	240	Dependent on events schedule			3rd Floor	1	0	1.92	Dran
Personal Computer	3	115	8 hours (5 days per week)	8	22	Treasury Department	3	0	60.72	Dran
Epson Printer	4	10	2 hours (3 days per week)	2	14	Treasury Department	4	0	1.12	Dran
Refrigerator	1	32	24 hours (5 days per week)	24	22	Treasury Department	1	0	16.896	Dran

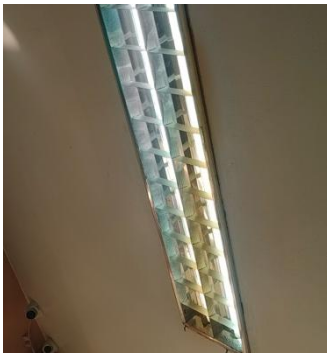
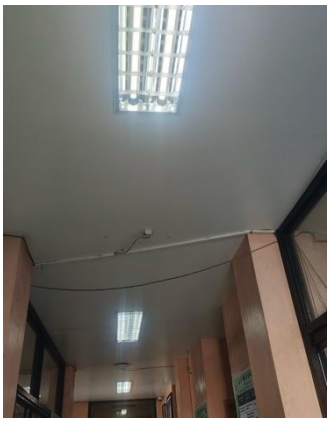
b) Energy Performance Analysis / Findings

i. Breakdown of Present Energy Consumption

Breakdown of Total Computed Energy Consumption	
Fixture	**Monthly Consumption (kWh)**
Lighting	265.352 kWh
HVAC	1824.908 kWh
Auxiliary	2947.367 kWh
Estimated Total	**5037.627 kWh**
Average kWh for 2023	**5169.92 kWh**
Percent Error $\frac{Actual - Expected}{Expected} x\ 100$	**2.625%**

c) Evaluation of Improvement Opportunities

Observation	Recommendation	Pictorial Evidence
Some doors and windows have gaps/are opened that can lead to inefficiency in air conditioning.	Keep the open spaces closed if there is continuous consumption of air conditioner	 

		
Most of the lights are still fluorescent type	Switch to LED lights as they consume less energy and power	 

These water leaks can cause electrical hazards especially on nearby appliances	Immediate repair and analysis of water leaks is needed to reduce the risk of affecting nearby electricity system	 
Glass windows and doors that leads outside are not tinted	Tinted windows are recommended to reduce significant increase in room temperature	 

d) **Recommendations for Improvement of Energy Performance**

i. Lighting Systems

Switching from fluorescent to LED lighting is a beneficial move, primarily due to LEDs' superior energy efficiency, which reduces electricity consumption by about 30% to 40% and lowers operating costs significantly. LEDs also boast a longer lifespan, reducing replacement and maintenance frequency. They provide better light quality, with more consistent output and no flickering or humming, enhancing visual comfort in various settings. Environmentally, LEDs are a more sustainable choice as they emit less heat, reducing cooling needs, and contain no mercury, easing disposal and reducing hazardous waste [7].

Upon replacing the 53 fluorescent lights in Barangay Lower Bicutan with LED lights, the estimated monthly energy consumption is projected to decrease to 120.872 kWh. This represents a reduction of 144.48 kWh compared to the estimated consumption if the fluorescent lights were retained, highlighting the total energy savings achieved through this upgrade.

ii. Cooling Devices:

1. Replacing a high-wattage fan, such as a Tornado fan consuming 350 watts per hour, with a more energy-efficient ceiling fan can lead to significant energy savings. Ceiling fans typically consume between 15 to 75 watts per hour, depending on their size and speed settings. For instance, a standard 48-inch ceiling fan might consume around 15 to 30 watts on low speed and 30 to 50 watts on medium speed.

   By replacing the Tornado fan with a ceiling fan, the estimated monthly energy consumption decreased from 1824.908 kWh to 1775.628 kWh. This adjustment results in a total energy savings of 49.28 kWh per month, highlighting the significant reduction in electricity usage achieved through the use of a more energy-efficient ceiling fan. This change not only reduces

operational costs but also contributes to more sustainable energy consumption practices [7].

2. Replacing conventional air conditioners with inverter-type systems is an effective strategy to enhance energy efficiency and reduce environmental impact. Unlike traditional models that operate at fixed speeds, inverter air conditioners adjust their compressor speed to match cooling needs, resulting in energy savings of up to 30% compared to non-inverter units [8].

By replacing the Carrier Aircon and Samsung Aircon units with inverter-type air conditioners, the total energy consumption was reduced from 18,524.908 kWh to 17,504.54 kWh, resulting in a notable energy savings of 1,020.368 kWh. This reduction highlights the superior efficiency of inverter technology, which adjusts compressor speed to match cooling needs.

iii. Auxiliary Systems

Devices that are plugged in but not in use such as Televisions, Rice Cooker, and Personal Computers still consume electricity (known as the phantom effect), unplugging those devices if not in use can save electricity [9].

On average, standby power for these devices can range from 1 to 10 watts. Assuming these devices are left plugged in 24 hours a day, this can amount to approximately 87.6 kWh per year for a television, 43.8 kWh per year for a rice cooker, and 28 kWh per year for a personal computer [10].

iv. Summary of Energy Savings Recommendations

Number	Energy Saving Recommendation	Annual Energy Saving (kWh/month)	Annual Savings	Capital Investment	Simple Payback Period
1	Replace 53 Fluorescent Tubes with LED Tubes	144.48 kWh/month	Php 19,067.89	Php 10,600	7 months
2	Replace the 2 Tornado BLASTER fan into Ceiling Fan	49.28 kWh/month	Php 6,503.78	Php 3,000	6 months
3	Replace Non-Inverter with Inverter Air Conditioner	1020.368 kWh/month	Php 134,664.09	Php 150,000	1 year and 1 month
4	Unplugging devices if not in use like Televisions, Rice Cooker and Personal Computer	6.713 kWh/month	Php 885.95	0	0
TOTAL		**1220.841 kWh/month**	**Php 161,121**	**Php 178,000**	

e) Cost-Savings Analysis

i. Formula Used

f) Monthly Energy Consumption (kWh)

Calculates the actual consumption of electricity by the building for each month.

$$MEC = \Sigma[Energy_{kW} \times t_{h/day} x\, t_{d/m}]$$

Where:

$Energy_{kw}$: Power Rating of Equipment (kW)

$t_{h/day}$: Hours of Use per Day

$t_{d/m}$: Days of Use per Month

g) Annual Energy Saving

Represents the amount of energy saved over a year as a result of implementing energy saving measures.

$$AES = \text{[Current MEC (kWh) – Estimated MEC (kWh)]}$$

where:

$Current\ MEC$: current monthly consumption

$Estimated\ MEC$: estimated monthly consumption after applying the energy saving measures

h) Annual Savings

Calculates the monetary savings due to the reduction in energy usage over one year.

$$U = 12 \times AES\ \left(\frac{kWh}{month}\right) \times Energy\ Rate\ \left(\frac{Php}{kWh}\right)$$

Where:

AES: Annual Energy Savings in (kWh/month)

Energy rate: Energy rate per kilowatt-hour, as charged by the energy provider

i) Simple Payback Period

Determines how long it will take for the initial cost of an energy-saving measure to be recovered, in years.

$$n_{spb} = \frac{Investment}{Annual\ Savings} = \frac{Investment}{U} = \left(\frac{\frac{Php}{Php}}{Yr}\right) = Yr$$

Where:

***Investment*:** Total Cost of proposed solution

U: Annual Savings

i. Project Energy Savings and Cost Reductions

Option A - Immediate		
	Average Monthly Consumption (kWh)	**Total Cost**
Current Monthly Energy Consumption	5169.92 kWh	Php 56,858.78
Monthly Consumption with Recommendation	5163.237 kWh	Php 37,790.89
Total Energy Saved	6.713 kWh	Php 73.83
Option B – Low Investment		
Current Monthly Energy Consumption	5169.92 kWh	Php 56,858.78
Monthly Consumption with Recommendation	4969.477	Php 54654.31
Total Energy Saved	200.473 kWh	Php 2204.80
Option C – High Investment		
Current Monthly Energy Consumption	5169.92 kWh	Php 56,858.78
Monthly Consumption with Recommendation	3949.109 kWh	Php 43,432.30
Total Energy Saved	1220.841 kWh	Php 13,426.81

Three potential strategies have been assessed to reduce electricity consumption and lower monthly energy costs. The option A, unplugging unused appliances**,** is a simple yet effective approach that can save approximately 6.713 kWh of energy each month, translating to a cost reduction of Php 73.83. This strategy requires minimal effort and offers immediate benefits in cutting down unnecessary energy usage.

The option B builds on the first by incorporating appliance upgrades alongside the practice of unplugging unused devices. Specifically, this involves replacing traditional fluorescent lights with energy-efficient LED tubes and swapping out a tornado fan for a ceiling fan. Together, these measures can achieve more substantial monthly energy savings of 200.43 kWh**,** resulting in cost savings of Php 2,204.80**.**

Finally, the option C combines all the actions from the first two strategies and further includes replacing a non-inverter air conditioner with an inverter model. This comprehensive approach offers the most significant savings, reducing energy consumption by 1,220.841 kWh each month and lowering electricity costs by Php 13,426.81. By implementing these strategies, households can make meaningful reductions in both energy usage and expenses, contributing to greater sustainability and cost-efficiency.

E. CONCLUSION

a. Summary of Findings and Recommendations

i. Key takeaways from the audit

- Appreciate the significance of assessing the energy efficiency of each appliance to minimize energy cost fluctuations.
- Building maintenance plays a significant role in enhancing the energy efficiency of appliances and lighting within the building.
- Unplugging the unused appliances can significantly reduce power energy consumption

ii. Emphasis on critical recommendations and their impacts

- Frequent site inspections of the deployed CCTV cameras are important to maintain the efficiency of the CCTV system.

Next Steps

iii. Suggested actions for the organization

- Prioritize transitioning from fluorescent to LED lights while optimizing their usage. Other appliances, especially the cooling system, should be maintained using inverters and ceiling fans.

iv. Monitoring and verification plan for implemented measures

- Implement a structured maintenance schedule for appliances with high energy consumption, such as CCTV cameras and cooling devices.

F. REFERENCES

[1] "Taguig City Profile – PhilAtlas," May 01, 1990. https://www.philatlas.com/luzon/ncr/taguig.html

[2] "Corporate profile," *Meralco*. https://company.meralco.com.ph/corporate-profile

[3] "Start or modify service," *Meralco*. https://company.meralco.com.ph/faq-about/start-or-modify-service

[4] "Forecasting: Principles and Practice (3rd ed)." https://otexts.com/fpp3/?

[5] S. J. T. & B. Letham, "Forecasting at scale," *ideas.repec.org*, 2018, [Online]. [1] Available: https://ideas.repec.org/a/taf/amstat/v72y2018i1p37-45.html

[6] "Forecasting : principles and practice : Hyndman, Rob J., author : Free Download, Borrow, and Streaming : Internet Archive," *Internet Archive*, 2014. https://archive.org/details/forecastingprinc0000hynd/page/8/mode/2up?.com=

[7] PacLights, "Benefits of retrofitting fluorescent lights to LED lighting fixtures," *PacLights*, Aug. 27, 2024. https://www.paclights.com/learning-center/benefits-of-conversion-from-fluorescent-lights-to-led/

[8] D. Hase, "Understanding ceiling fan electricity usage," *Kennedy Productions*, Aug. 24, 2023. https://www.kennedyproductions.com/fans/ceiling-fan-power-consumption/?.com

[9] "Does unplugging appliances save electricity? | Inspire Clean Energy." https://www.inspirecleanenergy.com/blog/sustainable-living/does-unplugging-appliances-save-electricity

[10] G. M, "Energy Vampires: How devices drain power when not in use," *Consumer Energy Center*, Jan. 02, 2025. https://www.consumerenergycenter.org/energy-vampire/

G. APPENDICES

a) Code

```
# !pip install prophet

# Import necessary libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from prophet import Prophet

# Load the dataset
df = pd.read_csv('Brgy-LowerBicutan-Monthly-Consumption.csv')

# Define your holidays as lists
holiday_dates = [
'2025-01-01', '2025-01-29', '2025-04-17', '2025-04-18', '2025-04-19', '2025-04-09',
'2025-05-01', '2025-06-12', '2025-08-21', '2025-08-25', '2025-11-01', '2025-11-30',
'2025-12-08', '2025-12-25', '2025-12-30', '2025-12-31', '2025-03-30', '2025-06-06',
'2025-07-27', '2025-10-31', '2025-12-24', '2026-01-01', '2026-02-17', '2026-04-02',
'2026-04-03', '2026-04-04', '2026-04-09', '2026-05-01', '2026-06-12', '2026-08-21',
'2026-08-31', '2026-11-01', '2026-11-30', '2026-12-08', '2026-12-25', '2026-12-30',
'2026-12-31', '2026-03-20', '2026-05-27', '2027-01-01', '2027-02-06', '2027-03-25',
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'2027-08-30', '2027-11-01', '2027-11-30', '2027-12-08', '2027-12-25', '2027-12-30',
'2027-12-31', '2027-03-09', '2027-05-16', '2028-01-01', '2028-01-26', '2028-04-13',
'2028-04-14', '2028-04-15', '2028-04-09', '2028-05-01', '2028-06-12', '2028-08-21',
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'2033-12-31', '2033-01-02', '2033-12-23', '2033-03-11', '2034-01-01', '2034-02-19',
'2034-04-06', '2034-04-07', '2034-04-08', '2034-04-09', '2034-05-01', '2034-06-12',
'2034-08-21', '2034-08-28', '2034-11-01', '2034-11-30', '2034-12-08', '2034-12-25',
'2034-12-30', '2034-12-31', '2034-12-12', '2034-03-01'
```

```
]

holidays_df = pd.DataFrame({
    'holiday': ['PH Holiday'] * len(holiday_dates),
    'ds': pd.to_datetime(holiday_dates)
})

# Function to align holidays to the first of each month
def align_holidays_to_month(holidays_df):
    holidays_df['ds'] = holidays_df['ds'].apply(lambda x: x.replace(day=1))
    return holidays_df.drop_duplicates()  # Drop duplicates if multiple holidays fall on the same month

# Apply the function to align holidays
holidays_df = align_holidays_to_month(holidays_df)

# Display the aligned holidays
print(holidays_df)

# Simulate an upward trend in the dataset
if 'y' in df.columns:
    df['trend'] = np.linspace(start=1, stop=1.5, num=len(df))  # Linear growth trend
    df['y'] = df['y'] * df['trend']  # Apply the trend to the original values
    df.drop(columns=['trend'], inplace=True)  # Drop the trend column to keep it clean

# Scale the historical data to keep the forecast
if 'y' in df.columns:
    max_value = df['y'].max()
    scaling_factor = 5000 / max_value  # Scale to a maximum of ~9,000 to allow for growth
    df['y'] = df['y'] * scaling_factor  # Apply scaling

# Initialize the Prophet model
m = Prophet(
    holidays=holidays_df,
    seasonality_mode='additive',
    changepoint_prior_scale=0.05,
    seasonality_prior_scale=0.1,
    holidays_prior_scale=10,
    yearly_seasonality=True
)

# Prepare the dataset for Prophet
df['ds'] = pd.to_datetime(df['ds'])  # Ensure the date column is in datetime format
m.fit(df)

# Generate future dataframe
future = m.make_future_dataframe(periods=132, freq='MS')  # Monthly forecast
forecast = m.predict(future)
```

```
# Display forecast tail
print(forecast[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].tail())

# Plot the forecast
fig1 = m.plot(forecast)
fig1.set_size_inches(12, 6)
x_ticks = pd.date_range(start=forecast['ds'].min(), end=forecast['ds'].max(), freq='YS')  # Year start
plt.xticks(
    ticks=x_ticks,
    labels=[tick.strftime('%Y') for tick in x_ticks],
    rotation=45,
    fontsize=10
)
plt.xlabel('Year', fontsize=12)
y_min = forecast['yhat'].min()  # Minimum forecasted value
y_max = forecast['yhat'].max()  # Maximum forecasted value
y_ticks = np.arange(
    start=500 * np.floor(y_min / 500),
    stop=500 * np.ceil(y_max / 500) + 500,
    step=500
)
plt.yticks(ticks=y_ticks, fontsize=10)
plt.ylabel('Energy Demand (kWh)', fontsize=12)
plt.grid(True, which='both', linestyle='--', linewidth=0.5)
plt.title('10-Year Energy Forecast (2025-2035) with Trend and Seasonality', fontsize=14)
plt.savefig('10-Year_Energy_Forecast_2025-2035.jpg', format="jpg", dpi=300, bbox_inches="tight")
plt.show()

# Plot the input data
plt.figure(figsize=(10, 6))
plt.plot(df['ds'], df['y'], label='Monthly Energy Consumption (2022-2024)', color='blue')
plt.title('Monthly Energy Consumption (2022-2024)')
plt.xlabel('Date')
plt.ylabel('kWh')
plt.xticks(rotation=90)
plt.legend()
plt.savefig("Monthly_Energy_Consumption_2022-2024.jpg", format="jpg", dpi=300, bbox_inches="tight")
plt.show()

# Plot the components of the forecast
fig2 = m.plot_components(forecast)
fig2.savefig("forecast_components.jpg", format="jpg", dpi=300, bbox_inches="tight")

# Create a table showing the peak kWh per year for the forecasted data (2025-2035)
forecast['year'] = forecast['ds'].dt.year
peak_kwh_per_year = forecast.groupby('year')['yhat'].max()  # Get peak values (max) for each year
```

```
peak_kwh_table = peak_kwh_per_year.reset_index()

# Display the peak kWh table
print("Peak kWh per year (2025-2035):")
print(peak_kwh_table)

# Save the peak kWh table to a CSV file
peak_kwh_table.to_csv("peak_kwh_output.csv", index=False)
```

b) **Documentation**

- **Clearance Room**



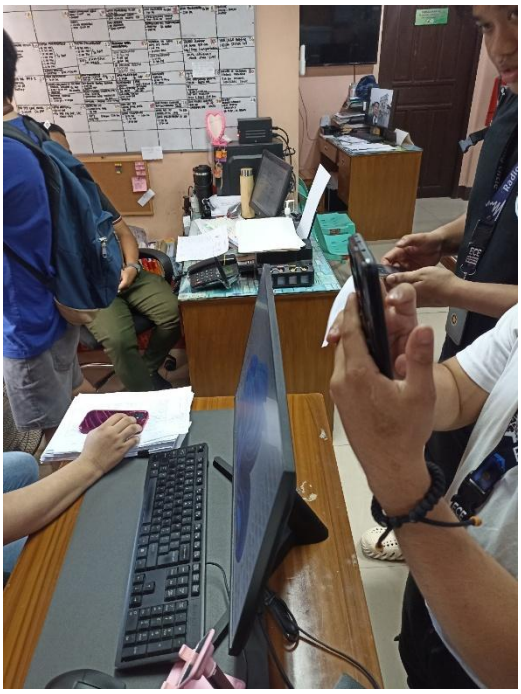


- **Annex Building and Barangay Security Force Room**





- **Room of the Secretary**



- **Office of Barangay Justice System Room and Office of Land Dispute Room**





- **Staircase to 3rd Floor and Session Hall**





- **3rd Floor and Treasury Department**



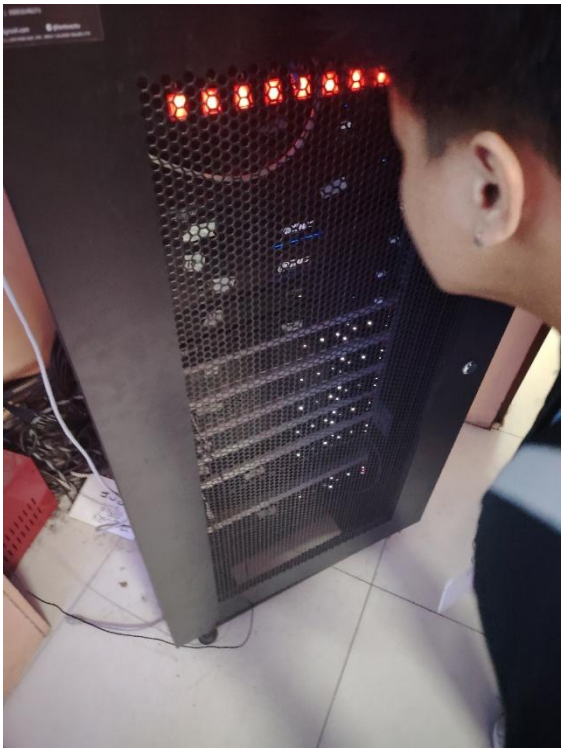


- **CCTV Room**









- **Stock Room of Treasury Department**





- **With Chairman Roel O. Pacayra**

